

SenecaBot — DeepMimic imitation-learning pipeline

Ovine (sheep) motion capture — C3D markers @ 200 Hz

MOCAP → reference-trajectory pipeline

seneca loco / simulation / mocap

io.py
C3D→markers

kinematics
markers→angles

build.py
FK→qpos/qvel

cyclic.py
cycle+symm+tile

traj_gen
site FK

*trajectory_adapted.npz — root pose + 12 joint trajectories
+ “*_mimic” site targets, resampled to the control rate*

Robot model — **senecabot_loco.xml**

12 DoF (4×3 hinges, sagittal plane) · direct-torque motors (DefaultControl)

Imitation training — Hydra + PPO

seneca_loco/.../train.py + conf.yaml! · loco-mujoco-seneca (fork)

ImitationFactory

TrajectoryHandler
(RSI)

GoalTrajMimic

★
MimicReward

★
PPOJax

MetricsHandler
(DTW / Fréchet)

Trained policy — **PPOJax_saved.pkl** (best-validation checkpoint)

Evaluation & diagnostics

eval.py (replay) · debug_agent.py (figures)
agent_diagnostics.py (speed_ratio, RMS, composite)

★ Local fork modifications (loco-mujoco-seneca)

MimicReward (reward/trajectory_based.py)

- rootvel — forward-velocity tracking term
- track_root_xy=false (drop unobservable root x,y)
- qvel_w_exp made tunable

PPOJax (algorithms/ppo_jax.py)

- live wandb logging (io_callback)
- best-checkpoint buffer
- live validation streaming

MujocoViewer (visuals)

- record_camera — video from a named XML camera

Additive & opt-in;
no change to PPO math.